

Algebra

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Chapter 1

Groups

1.1 Introduction

Definition 1 (Group). A **Group** $(G, \cdot)$ is a set G which has an operation $\cdot : G \times G \rightarrow G$ that is:

1. $f \cdot (g \cdot h) = (f \cdot g) \cdot h \quad \forall f, g, h \in G$ (**Associativity**)
2. $\exists e \in G$ t.c. $g \cdot e = e \cdot g = g \quad \forall g \in G$ (**Neutral Element e**)
3. $\forall g \in G \exists g^{-1}$ t.c. $g^{-1} \cdot g = g \cdot g^{-1} = e$ (**Inverse Element**)

Note 1. There's the Associativity property, so I can always write:

$$f \cdot g \cdot h = (f \cdot g) \cdot h = f \cdot (g \cdot h)$$

I'll also omit the dot when it is clear in the context.

Definition 2. A group is called:

1. An **Abelian Group** if $f \cdot g = g \cdot f \quad \forall f, g \in G$
2. A **Finite Group** if $|G| < \infty$

Definition 3 (Subgroup). Let $(G, \cdot)$ be a group. Let $H \subseteq G$ be a subset. H is a **G-Subgroup** if $(H, \cdot)$ is a group.

Proposition 1. Let G be a group.

Let H t.c. $|H| < \infty$ a G subset. It is a G -subgroup if it is closed in the group operation.

Proof. We need to demonstrate that $(H, \cdot)$ is a group.

Let's start by assuming that H is finite and closed in the group operation.

- $\forall g \in H, \quad g \in G \Rightarrow$ Associativity is granted.
- $|H| < \infty \Rightarrow \text{grad}(g) = n_g \Rightarrow g^{n_g} = e \Rightarrow e \in H, \quad g^{-1} = g^{n_g-1} \in H$

□

1.2 Lagrange Theorem

Definition 4 (Congruent in module). Let G be a group, $H \leq G$ a subgroup, $f, g \in G$. f is called **congruent to $g \bmod H$** if:

$$f^{-1} \cdot g \in H$$

In this case I'll write: $f \equiv g \pmod{H}$ or $f \equiv_H g$

Proposition 2. The property of being congruent in module is a class of equivalence.

Proof. Exercise. □

Proposition 3 (Class of equivalence). Let $(G, \cdot)$ be a group, $H \leq G$ a subgroup, $g \in G$. Therefore the class of equivalence of g in H is:

$$gH := \{g \cdot h \mid h \in H\}$$

Proof. Exercise. □

Definition 5. The set of gH for all $g \in G$ is written: G/H

Definition 6. let a group be G and $H \leq G$ a subgroup. H is called **Normal** if: $gH = Hg \quad \forall g \in G$. In that case It's written: $H \trianglelefteq G$.